



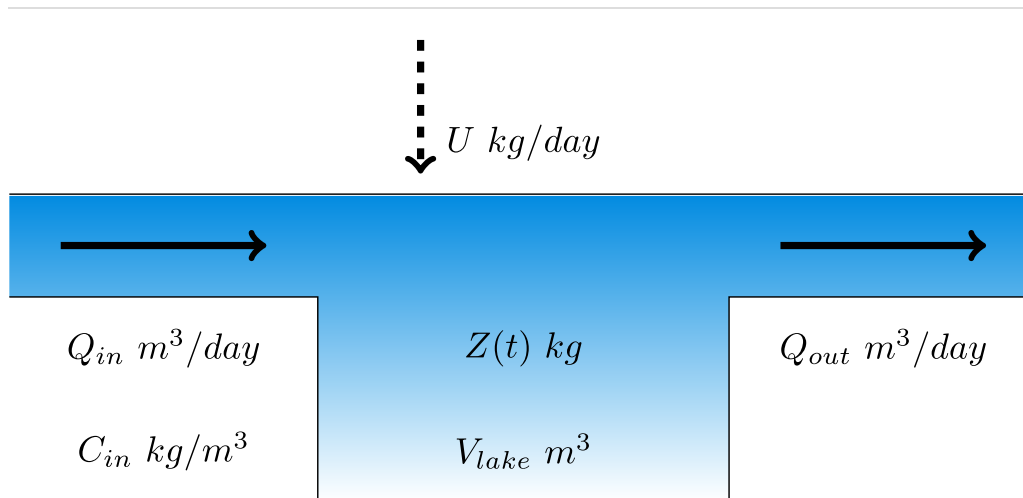
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Mathematical model

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In the last section we formulated the mixing problem, and defined the relevant quantities. The question we want to answer is:

A factory wants to start discharging waste material in a lake. The river that feeds into the lake is already polluted somewhat. How much would this new contamination increase the pollution in the lake?



We want to calculate Z , the mass of the pollutant in the lake. The relevant quantities are shown in the schematic above.

Which quantities are constant?

We are investigating the change caused by the factory, so the simplest case is then to take all other variables constant.

Take C_{in} constant. Assume also that Q_{in} and Q_{out} are equal and constant in time, so just name both Q : $Q_{in} = Q_{out} \equiv Q$. With equal flow rates in and out, V_{lake} is also constant (assuming that the volume of the discharge of the factory is negligible).

To see what influence the discharge of the factory has, we will assume that before $t = 0$ the factory is not yet polluting, so $U = 0$ for $t < 0$, and that $U > 0$ for $t \geq 0$. So U does change in time. The nice thing is however, that our calculations start from $t = 0$, so in our calculations U can be a constant as well!

Without the discharge of the factory, the pollution concentration in the lake is (after some time) the same as the concentration in *River_{in}*: $Z/V_{lake} = C_{in}$. So we will take that as the initial condition for Z :

$$Z(0) = C_{in} V_{lake}.$$

Define Δt as the length of a small time interval $[t, t + \Delta t]$.

Define ΔZ as the change in Z during that time interval, so

$$\Delta Z = Z(t + \Delta t) - Z(t).$$

Construct the balance equation for ΔZ . The expression for ΔZ should have three terms: one for the inflow, one for the factory discharge and one for the outflow.

If you are done, check that the unit of each of the terms is the same as the unit for ΔZ .

Exercise A

1/1 point (graded)

The balance equation for ΔZ is:

☒ $\Delta Z = + C_{in} Q \Delta t + U \Delta t - \frac{Z(t)}{V_{lake}} Q \Delta t$ ✓

☐ $\Delta Z = + C_{in} Q \Delta t + U \Delta t - C_{in} Q \Delta t$

☐ $\Delta Z = + C_{in} \Delta t + U \Delta t - \frac{Z(t)}{V_{lake}} \Delta t$

☐ $\Delta Z = + C_{in} Q + U - \frac{Z(t)}{V_{lake}} Q$

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You have used 1 of 2 attempts

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

Divide the correct balance equation by Δt and take the limit for $\Delta t \rightarrow 0$.

The initial value problem for $Z(t)$ is:

☒ $\frac{dZ}{dt} = -\frac{Q}{V_{lake}}Z(t) + C_{in}Q + U, \quad Z(0) = C_{in}V_{lake}$ ✓

☐ $\frac{dZ}{dt} = -\frac{Q}{V_{lake}}Z(t) + C_{in}Q + U, \quad Z(0) = C_{in}Q$

☐ $\frac{dZ}{dt} = +\frac{C_{in}}{V_{lake}}Z(t) + C_{in}V_{lake} - UQ, \quad Z(0) = C_{in}V_{lake}$

☐ $\frac{dZ}{dt} = +\frac{C_{in}}{V_{lake}}Z(t) + C_{in}V_{lake} - UQ, \quad Z(0) = 0$

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You have used 1 of 2 attempts

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In the next section, you will investigate the solution of this initial value problem.

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